HOW TO LOCATE A DEVICE Without Using GPS?

SSID-based Wi-Fi locationing offers a precise method for determining locations using Wi-Fi signals, particularly in areas where GPS often proves unreliable.

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A new approach to location tracking is redefining how devices can be found when traditional methods like GPS and cellular systems fall short. SSID (service set identifier)-based Wi-Fi locationing excels in places where GPS struggles, such as dense urban areas and indoor settings, delivering a rare combination of precision and energy efficiency.

Rather than relying on satellite signals or cell towers, SSID-based Wi-Fi locationing leverages signals from nearby Wi-Fi access points to determine a device's location. It achieves a noteworthy balance, offering higher accuracy than cellular systems, albeit with slightly less precision than GNSS. Its real strength lies in its energy efficiency—consuming less power than GNSS while maintaining performance on par with cellular systems.

This technology is already integrated into many modern devices, including smartphones, where it enhances real-time tracking in difficult conditions. SSID-based locationing works seamlessly with other data inputs, such as compass metrics, to provide reliable and comprehensive location tracking. Its practical applications are far-reaching, from indoor navigation in GPS-obstructed environments to asset tracking in large facilities like warehouses or factories, and personnel monitoring in venues or events. These capabilities not only improve security and operational efficiency but also enable targeted engagement through location-based services, including personalised advertising and marketing campaigns.

Wi-Fi locationing simplified

The process of Wi-Fi locationing utilises information such as the SSID, BSSID (basic service set identifier), and the signal strength from nearby networks. This data is then compared against a pre-existing database of Wi-Fi networks that have known geographical locations. The ability to pinpoint a device's location through Wi-Fi significantly enhances